

The University of Nottingham

SCHOOL OF Computer Science

A LEVEL THREE MODULE, SPRING SEMESTER 2017-2018

Computer Vision

Time allowed TWO HOURS

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer THREE Out of FOUR Questions

Each question carries 20 marks

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: None

INFORMATION FOR INVIGILATORS: None

1. The following questions are pertaining to Features and Image Segmentation.

- a. Patch based features are widely used in Computer Vision to describe certain characteristics of an image patch. To generate patch based features, it generally involves extracting some basic feature for each pixel and summarising the basic features over all the pixels in the patch.
- i. Describe the basic features that are extracted at the pixel level in Local Binary Pattern (LBP) and Histogram of Oriented Gradients (HOG), respectively. [3 Marks]
 - ii. Histogram is often used for summarising basic features over all the pixels in an image patch. Are such summarised features robust to rotation and translation of the image? Explain. [2 Marks]
 - iii. In both LBP and HOG methods, histograms of basic features are not computed over the entire image but instead in sub-windows. How are the histograms in sub-windows used to form final features of the image? Why is that strategy adopted? Explain. [2 Marks]
- b. An image is shown in Figure 1. You are asked to design an approach to segment the image into different regions. You need to make choices on a feature representation for each of the pixels in the image and a segmentation method to assign each of the pixels into its desired region. Three types of feature representation are available for you to choose, including histogram of intensity, LBP, and HOG. In order to obtain the feature for each of the pixels, it is computed over a small window that is centred on the pixel. The two segmentation methods that you can select are K-means and Gaussian Mixture Models (GMM).



Figure 1

- i. The task is to segment the image into three regions representing the person, the ground covered by the snow, and the sky. Which of the three features will you choose? You should evaluate each of them and justify your decision. [6 Marks]
- ii. Compare and contrast the principles and algorithmic procedures of k-means and GMM for image segmentation tasks. [5 marks]
- iii. Assume that you will use the histogram of intensity to represent each of the pixels in the image. Out of k-means and GMM, which one will you choose to perform the segmentation task in (i)? Explain. [2 Marks]

2. The following questions are pertaining to Motion, Optical Flow, and Tracking.

- a. To evaluate the motion of moving objects, in both Horn-Schunk and Lucas-Kanade methods, the concept of optical flow is used and it must satisfy two constraints, i.e., brightness constancy and smoothness.
 - i. Explain how smoothness constraint is expressed in Horn-Schunk and Lucas-Kanade, respectively. [2 marks]
 - ii. Describe the concept of image pyramid and explain how it is used with Lucas-Kanade to overcome the issue caused by large motions. [6 marks]
 - iii. Motion field is not necessarily the same as optical flow. Give an example and explain how the motion field differs from the optical flow in your example. [2 marks]

- b. Suppose you are tracking the target that is highlighted by the bounding box in Figure 2. Particle filters use a set of samples to approximate the target probability density, where each sample is a particle, representing a hypothesised target state with some probability being correct. CONDENSATION is the original algorithm that implements particle filters for tracking targets.
- Assume a particle hypothesising the target, is centred at where it is marked with an "O" that is next to the bounding box. Explain how you may measure the weightage of this particle. Is the weightage to be likely high or low according to your way of measurement? Elaborate.
[4 marks]
 - CONDENSATION requires maintain the same number of particles throughout the tracking process. How is it done? Elaborate your answer with the image in Figure 2.
[3 marks]
 - What advantage does particle filters have over optical flow methods, e.g., Lucas-Kanade, for tracking targets in real world? Explain with the situation in the image in Figure 2.
[3 marks]

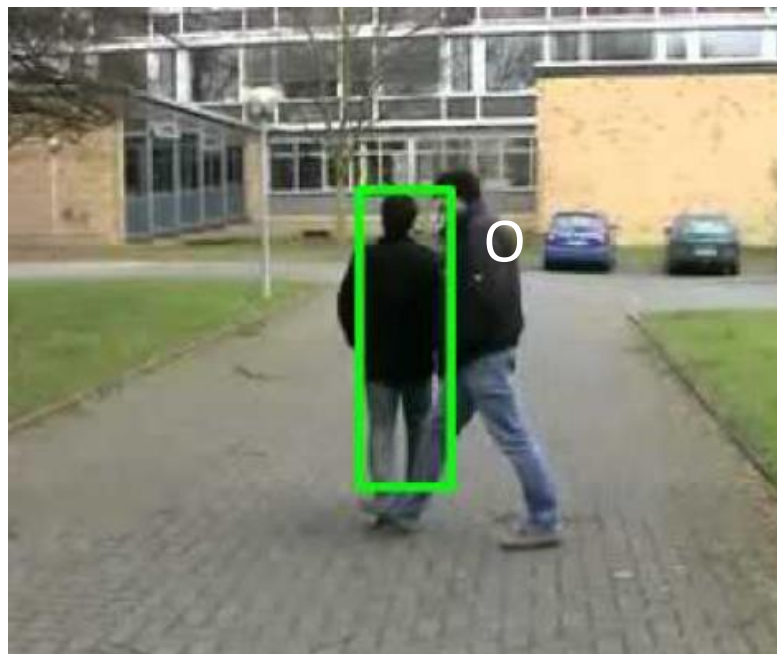


Figure 2

3. The following questions are pertaining to Stereo Vision.

- a. A typical stereo vision system usually consists of four stages, including camera calibration, image rectification, correspondence matching, and depth estimation.
 - i. Explain briefly the purposes of camera calibration and image rectification, respectively. [2 marks]
 - ii. Correspondence matching is governed by epipolar geometry. Sketch and explain the main components of and the principle behind the epipolar geometry. [4 marks]
 - iii. Patch based correlation method can be used to search the correspondences in the right image for the pixels in the left image in Figure 3. Discuss the effect of different patch window size (i.e., small vs. large) may have on searching the correspondences for the pixels in the regions marked with an "a" and a "b", respectively. [4 marks]



Figure 3

- b. Consider again the pair of rectified images in Figure 3 for stereo correspondence. Global methods represent dense correspondence between the two images as a continuous or discrete function defined over the coordinates of image pixels. For example, $d(x, y)$ is the disparity between the pixel at (x, y) in the left image and the pixel at $(x + d(x, y), y)$ in the right image. Matching correspondence is found to minimise a global cost function of image data and smoothness constraint.

- i. Explain briefly what aspects of the cost measurement the image data and smoothness constraint refers to, respectively. [2 marks]
- ii. Given that pixel A matches A' and pixel B matches B', the ordering constraint requires that if A is to the left of B in the left image, A' must be to the left of B' in the right image. Ordering constraint is often useful but may be violated under certain circumstances. Sketch an example and explain how the ordering constraint is violated in your example. [4 marks]
- iii. Dynamic programming can be used to find disparity along a particular scanline, where ordering constraint is normally assumed to be valid. An example of optimal path through grid is shown in Figure 4, where diagonal lines represent matched pixels between the left and right images, and horizontal or vertical lines represent unmatched pixels in either the left or right image, respectively. Using the style in Figure 4, sketch a possible path yielded by dynamic programming for the scanline highlighted as a dotted line in Figure 3. Your sketch should satisfy the ordering constraint, and reflect reasonably observable occlusions in left and right image respectively. Your sketch does not need to show the actual image width. Explain your sketch. [4 marks]

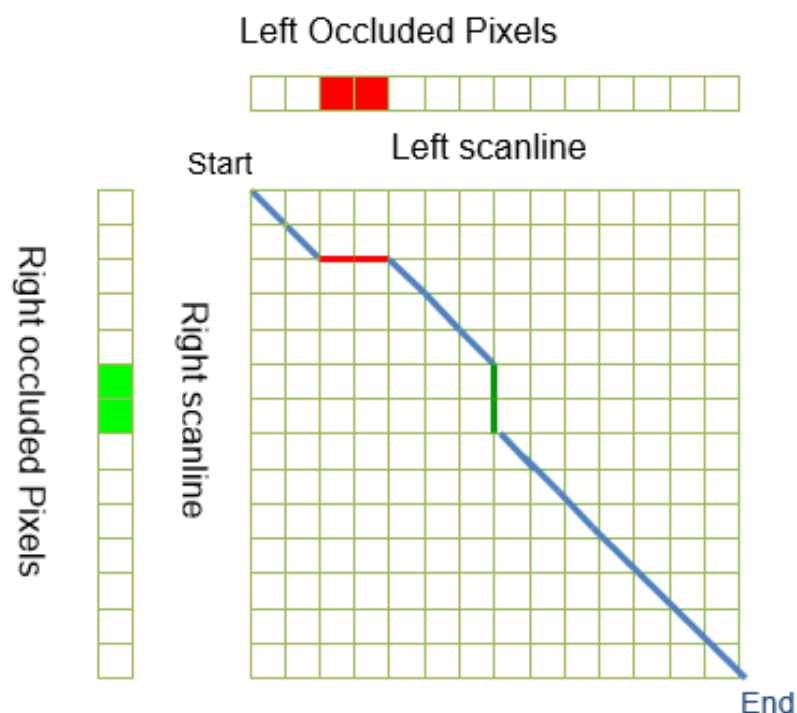


Figure 4

4. The following questions are pertaining to Features, Object Recognition, and Machine Learning approaches.

- a. Bag-of-Features is a feature representation technique that describes an image as a histogram of local image features. Each local image feature is analogous to a word in a document. Thus all different local image features would form a codebook, analogous to vocabulary.
- i. Describe the general process of constructing a codebook. [3 marks]
- ii. Usually the bag-of-features of an image is a sparse vector, i.e., only a few entries of the histogram of the codebook is non-zero. It is inefficient for the purpose of image retrieval in large image database. Inverted file technique can help alleviate the problem. Explain the concept of the inverted file and how it is used to improve the retrieval efficiency. [3 marks]
- b. Viola-Jones object detection framework provides competitive object detection in real time. It was originally designed for face detection. It involves mainly three aspects including integral image for fast feature extraction, boosting for feature selection, and attentional cascade for fast rejection of non-face windows.
- i. Four basic types of features are extracted from an image window that is under examination. Sketch the four basic types and describe how they are used in the Viola-Jones framework to extract larger set of features. [4 marks]
- ii. Usually not each of the features in a large set is useful or relevant for detecting whether a face is present in the image window. Explain how boosting is used to select relevant features. [4 marks]
- c. Convolutional Neural Nets (CNNs) have had a huge impact on computer vision in recent years. Briefly describe the components of a CNN for object recognition and show how they are connected to form a complete system. [6 marks]